

Adaptive Query-Specific Retrieval Optimization Using Contextual Bandits

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Outline

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- Approaches
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 - PPO
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Motivation: RL for Information Retrieval

Goal: Apply RL techniques to improve precision of **info retrieval** while optimizing query efficiency & balancing cost and performance.

- Consider **one-hop** retrieval, **single-turn query** setting.
 - Multi-hop or multi-turn could be modeled as MDP, but way more complex to deal with.
 - **Hybrid:** Dense retrieval (SBERT embeddings) + Sparse retrieval (keyword-based ranking with TF-IDF)
- Use policy-based retrieval optimization to decide:
 - What retrieval parameters to use: **Top-k** selection (small or large k), or **threshold & weights** of similarity score + BM25.
 - **Hyperparameters tuning of hybrid retrieval.**



Problem Definition

- Given a **Hybrid Retrieval System** combining BM25 and dense embeddings (SBERT), **cosine similarity score** between given query and documents.

Multi-Armed Bandit Problem:

- **Stateless** (MAB setting)
- **Action Space**
 - **Top-k selection:** $k \in \{1, 2, \dots, 10\}$. Controls how many top docs to retrieve.
 - **BM25 weight (λ):** $\lambda \in [0, 1]$, continuous or discrete (in increments of 0.1). Controls weight between sparse and dense retrieval, $\lambda = 1$ would lead to full sparse retrieval.
 - **Similarity threshold (τ):** $\tau \in [0.5, 0.8]$, continuous or discrete (in increments of 0.05). Controls the threshold that decides what docs should be returned based on the scoring.



Reward Function

- $R = \alpha \cdot \text{Retrieval Acc} - \beta \cdot \text{Token Cost} - \text{Penalty}$,
where we choose $\alpha = 1$ and $\beta = \frac{1}{2000}$.
- **Retrieval Accuracy:** Relevance score, calculated by Jaccard similarity (Maximum token overlap between retrieved documents and ground-truth document).
 - If ground truth is within the retrieved docs and pass the threshold check, the Acc score will be 1.
- **Token Cost:** Total token length of retrieved documents.
- **Penalty:** Additional penalty to encourage maintaining reasonable threshold values.
 - Low τ values will be punished to discourage overly loose similarity filtering.



Approaches

- **Non-Contextual MAB** as baseline
 - **Epsilon-Greedy** exploration with $\varepsilon = 0.05$.
 - The agent tries to learn the best retrieval action across all queries without context, trained once on the train set.
- **Contextual MAB**
 - Added the **query context features** (dense query embeddings, query length, token diversity, and first-word type), trained once with LinUCB on the train set.
 - **LinUCB**: standard CMAB algorithm that fits a linear model given the input context for each action.
 - PCA applied to reduce feature dimensionality.
- **Advantage Actor-Critic (A2C)**
 - Separate actor and critic heads share a common feature extractor.
 - The agent learns to predict **action distribution** and **value function** simultaneously.
 - Joint training of actor and critic heads for retrieval action prediction and value estimation.
- **Proximal Policy Optimization (PPO)...**



Main Approach Design - PPO

Agent Behavior:

- The agent samples actions from a hybrid policy:
 - **Discrete choice** for **top-k** selection
 - **Continuous** for τ (**similarity threshold**) and λ (**hybrid blending weight**).

Policy Network:

- The policy network consists of a shared two-layer MLP that encodes query features.
- Separate heads predict:
 - **Top-k logits.**
 - **τ values** and **λ values.**
- τ and λ are passed through a sigmoid activation to constrain their range to $[0, 1]$.



PPO Training Details

- **Clipping** applied to restrict ratio between new and old policy probabilities, **stable policy updates**.
 - Clip Threshold: 0.2
- **Clipped surrogate loss.**
- **Entropy bonus** added to encourage sufficient exploration.
 - Entropy coeff: 0.01
- **Normalization of the advantage** within each batch to stabilize learning.
- **Batch:** 256 queries, **Learning rate:** 1e-4,
- **Updates:** 5 optimization epochs per iteration, with 40 iterations in total.



Feature Inputs (LinUCB and PPO)

- **Dense query embeddings** – vector representation of the query (from SBERT). PCA applied for LinUCB.
- **Query length** – number of tokens, normalized by dividing by 10.
- **Token diversity** – ratio of unique tokens to total tokens.
- **First-word type** – 6 binary features indicating if the query starts with *what*, *who*, *when*, *where*, *why*, or *how*.



Data Preprocessing

- **Text Tokenization**

- Tokenized versions were used to build a BM25 index.
- Documents preprocessed by lowercasing and tokenizing into words using regular expressions for BM25 indexing.

- **Document and Query Encoding**

- Used **all-MiniLM-L6-v2** Sentence-BERT (SBERT) to encode documents and queries into dense embeddings.
- Embeddings normalized (L2) for improved cosine similarity.

- **FAISS flat L2 Index Construction**

- **Hybrid Retrieval Setup**

- **Batch Pre-encoding**



Dataset

- Sampled the first **5,300 examples** from the training split to form our retrieval corpus., with 5,000 for train set and 300 for test set
 - **SQuAD** (Stanford Question Answering Dataset)
 - **NQ** (Natural Questions)
- Each sample in the dataset contains:
 - A **question** (user query)
 - A **context** paragraph (reference document)
- For our retrieval setup:
 - Truncated each context to 500 characters to simulate more realistic retrieval passages.
 - **Questions** as the input queries for retrieval optimization experiments.



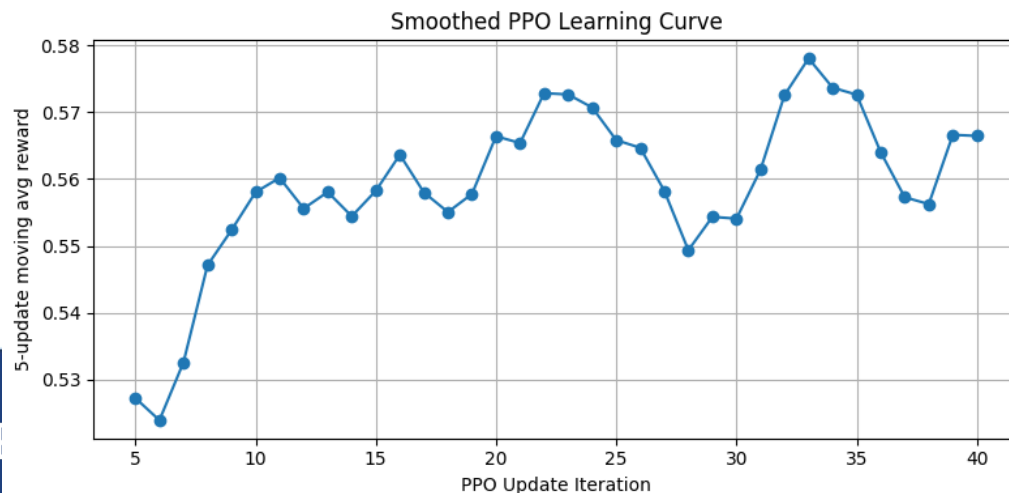
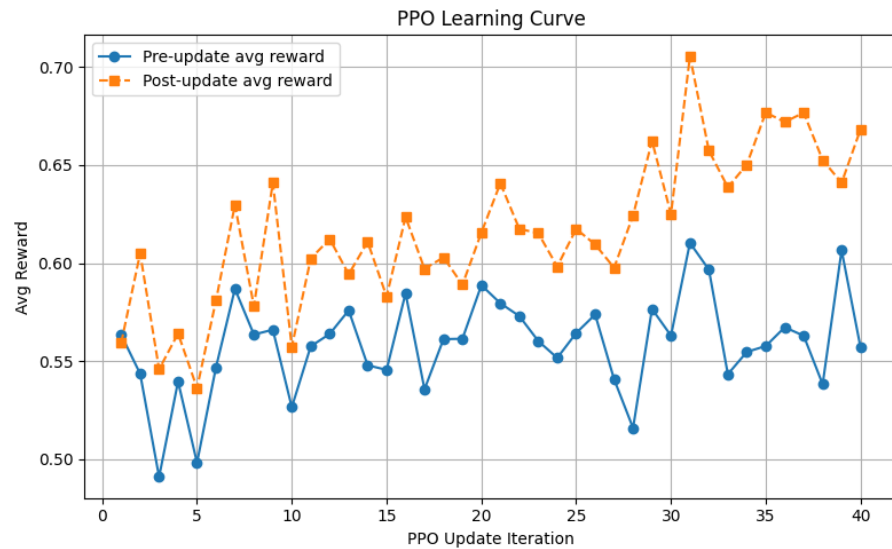
Experimental Results (Overall performance)

Dataset	SQuAD			NQ		
Metric Model	Avg Reward	Avg Relevance Score (Acc)	Avg Token Cost	Avg Reward	Avg Relevance Score (Acc)	Avg Token Cost
ϵ – Greedy (MAB)	0.5817	0.6067	48.9	0.8531	0.9064	105.17
LinUCB (CMAB)	0.5456	0.6333	158.72	0.9018	0.9667	115.1733
A2C	0.5868	0.7567	311.59	0.907	0.98	116.74
PPO	0.6345	0.71	135.5	0.9261	0.9967	119.99

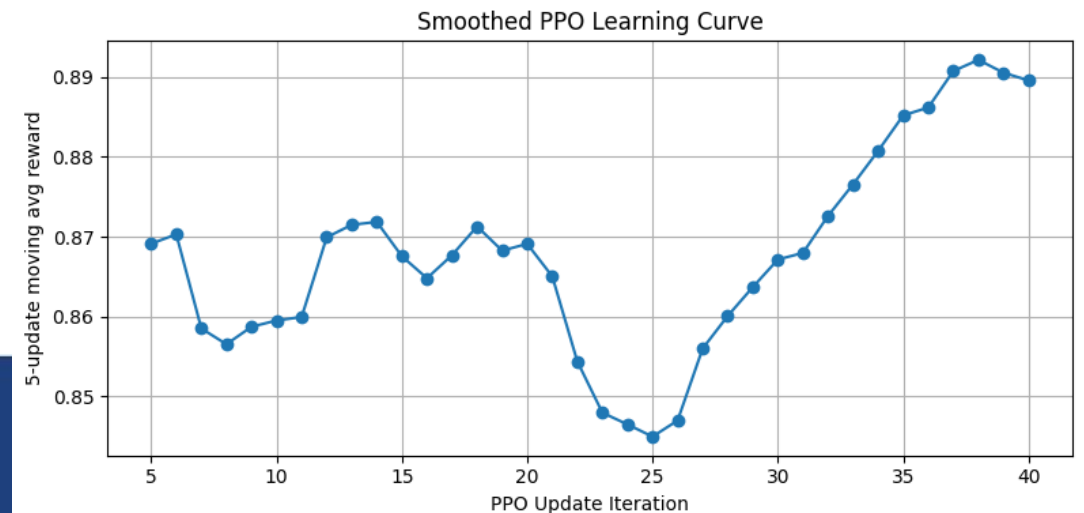
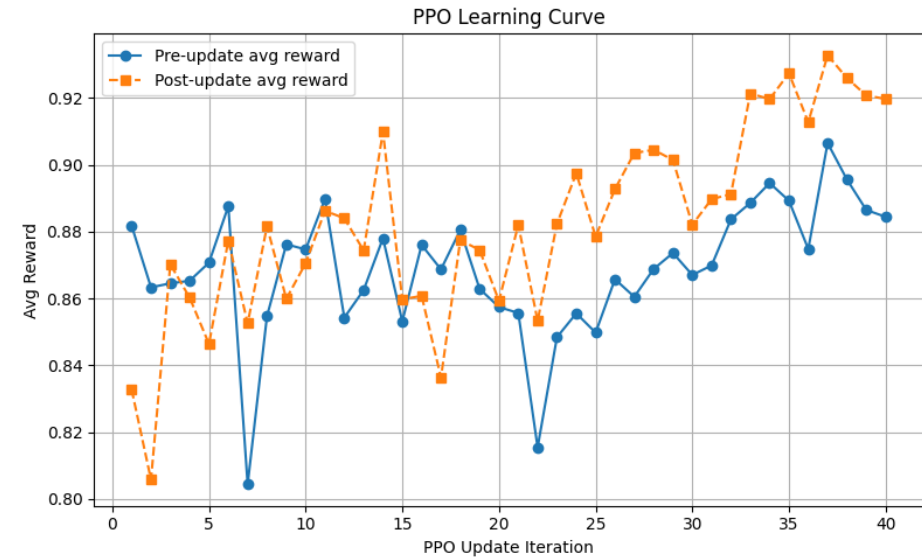


SQuAD v.s. NQ Learning Curves (PPO)

SQuAD



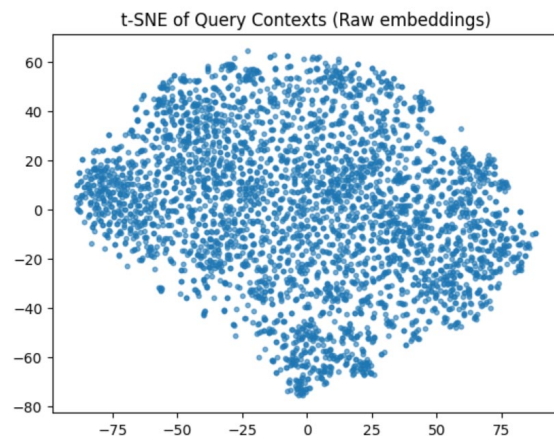
NQ



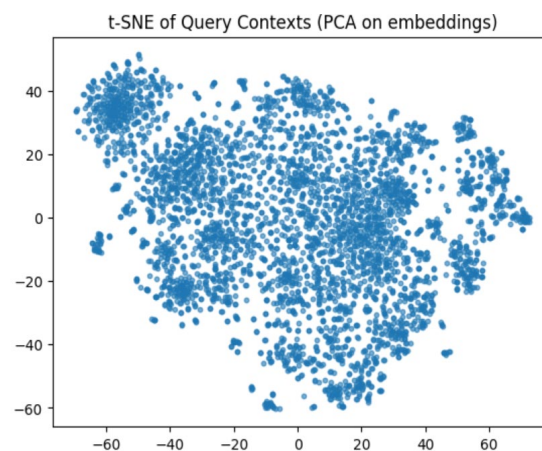
t-SNE of Query Contexts

NQ

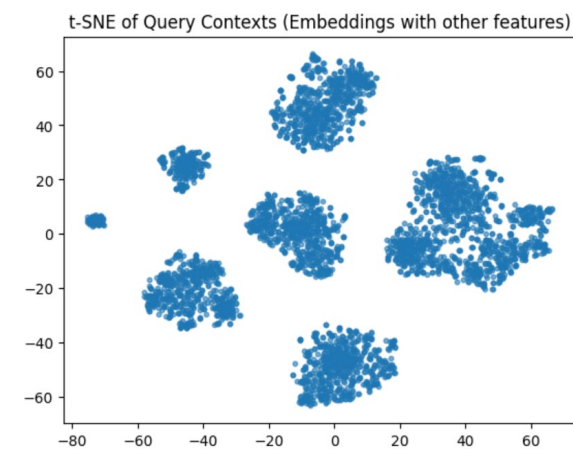
Raw embeddings



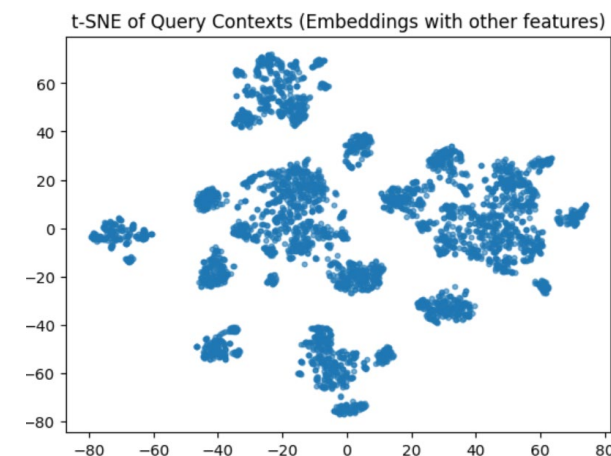
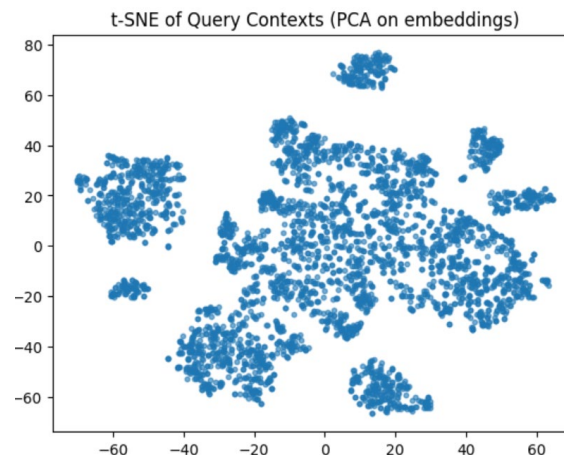
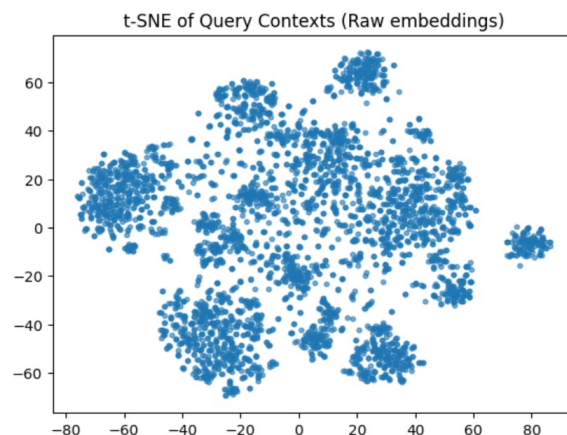
PCA



With other features



SQuAD

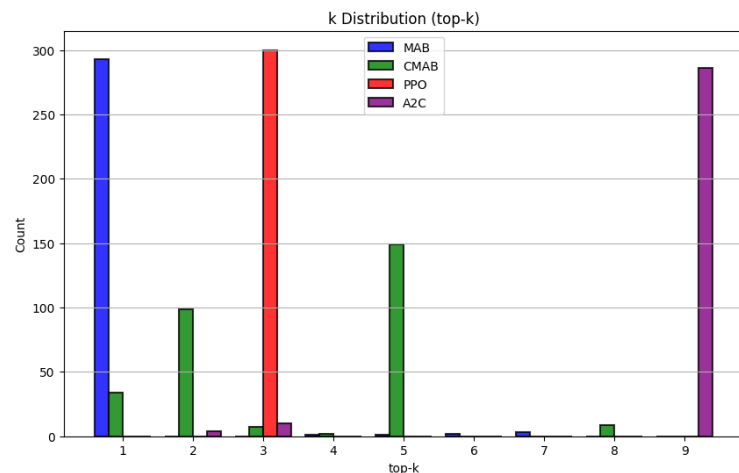


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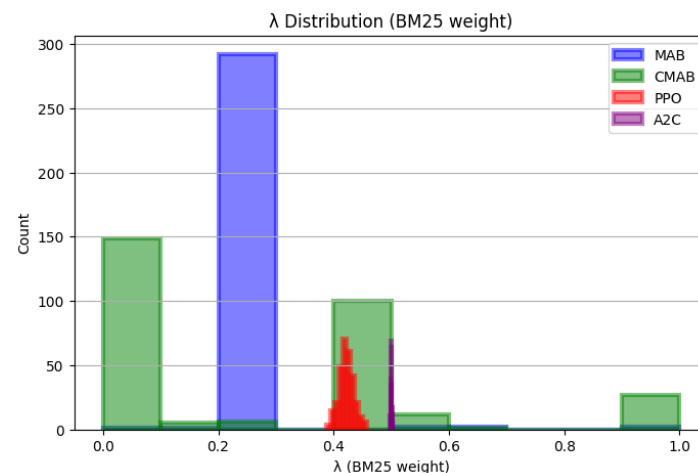
Distribution of Selected Retrieval Actions

SQuAD

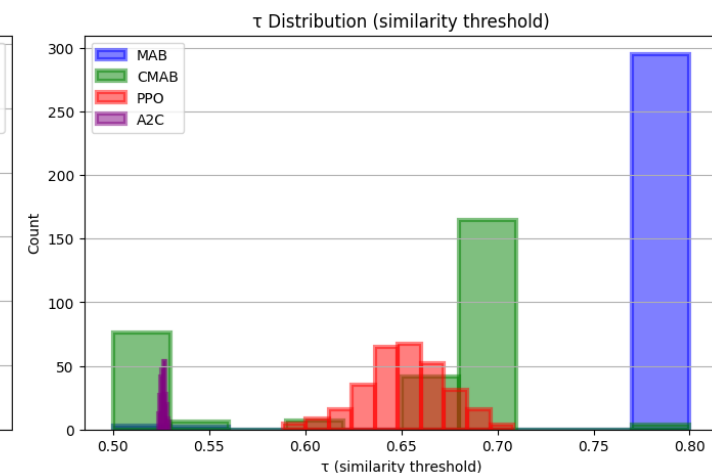
top-k



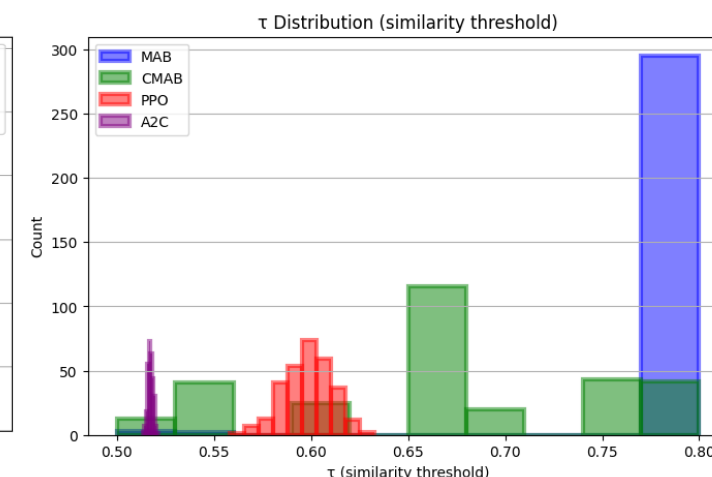
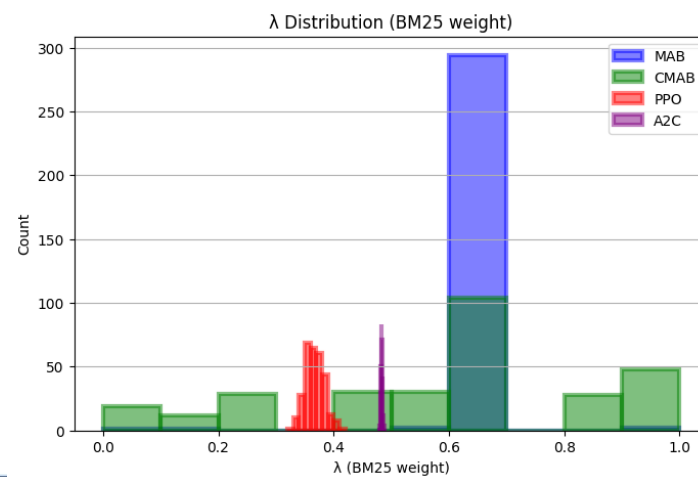
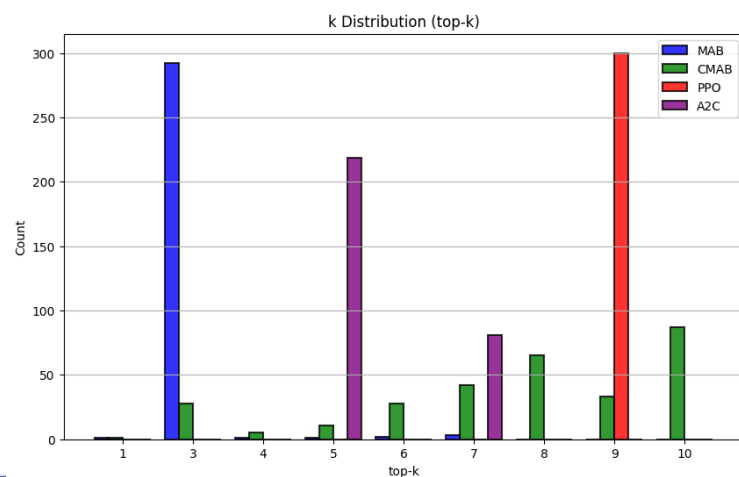
λ (lambda)



τ (tau)



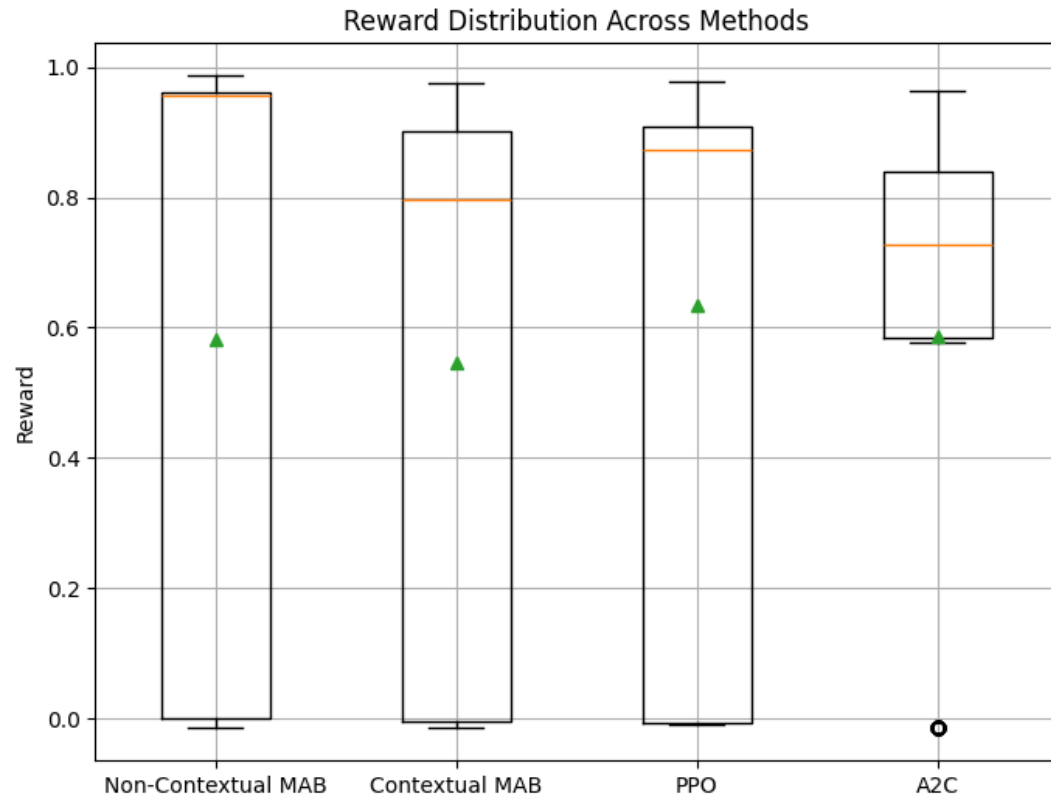
NQ



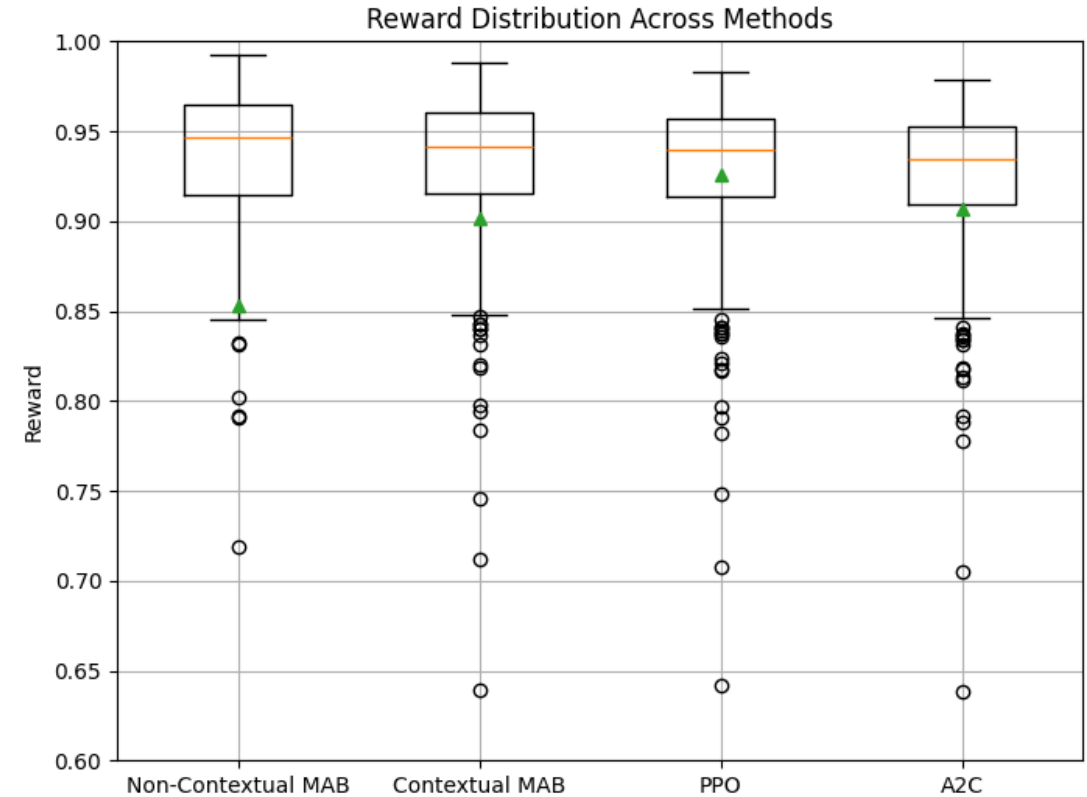
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Rewards Distribution

SQuAD



NQ



Conclusion

- **Contextual Models vs. Non-contextual Baselines:**
Contextual models can outperform non-contextual baselines.
- **Deep RL Methods:**
PPO achieves the best performance in balancing retrieval relevance and cost, though it requires more training time.
- **Feature Engineering:**
Query embeddings and handcrafted features (e.g., query length, token diversity) enhance model effectiveness.
- More analysis could be done...
- Show the potential of applying adaptive learning techniques for query-specific optimization in information retrieval systems.



Thank you!



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